Integrale definito

1 Calcolare i seguenti integrali

1.1) $\displaystyle\int_1^2 \left(\sqrt{x}+\frac{1}{x}+\frac{1}{x^2}\right)dx$ **1.2)** $\displaystyle\int_{-1}^2 \left(x^2-8x+x^3+\sqrt[3]{x}\right)dx$

1.3) $\displaystyle\int_1^2 \frac{6x^2+3x+4}{x}\,dx$ **1.4)** $\displaystyle\int_0^\pi (\cos x+\sin x)\,dx$

1.5) $\displaystyle\int_{-2}^3 (\sqrt{|x|}-\sqrt[5]{x-1})\,dx$ **1.6)** $\displaystyle\int_0^{\pi/2} (\cos x+\arctan x)\,dx$

1.7) $\displaystyle\int_{-1}^{\sqrt{3}} \frac{5}{1+x^2}\,dx$ **1.8)** $\displaystyle\int_0^1 \frac{x}{\sqrt{x+1}}dx$ **1.9)** $\displaystyle\int_0^{2\pi} |\cos x|\,dx$

1.10) $\displaystyle\int_0^3 ||x-2|+3|\,dx$ **1.11)** $\displaystyle\int_0^5 \{x\}\,dx$ **1.12)** $\displaystyle\int_{-2}^2 [x^2]\,dx$ **1.13)** $\displaystyle\int_0^8 \{\sqrt[3]{x}\}\,dx$

1.14) $\displaystyle\int_0^1 xe^{x^2}\,dx$ **1.15)** $\displaystyle\int_1^e x\log x\,dx$ **1.16)** $\displaystyle\int_0^{2\pi} \cos^2 x\,dx$ **1.17)** $\displaystyle\int_0^{\pi/2} \sin^3 x\,dx$

1.18) $\displaystyle\int_1^e \frac{\log^2 x}{x}\,dx$ **1.19)** $\displaystyle\int_1^4 x\sqrt{1+x^2}\,dx$ **1.20)** $\displaystyle\int_2^3 x^4(x^5+1)^2\,dx$

1.21) $\displaystyle\int_{\pi/6}^{\pi/3} \tan x\,dx$ **1.22)** $\displaystyle\int_0^{\pi/6} x\tan(x^2)\,dx$ **1.23)** $\displaystyle\int_0^{\pi/4} \sin^3 x\cos x\,dx$

1.24) $\displaystyle\int_1^e \frac{1}{x(3+\log x)^2}\,dx$ **1.25)** $\displaystyle\int \frac{1}{\sqrt{1-4x^2}}\,dx$ **1.26)** $\displaystyle\int \frac{2x}{\sqrt{1-3x^4}}\,dx$

1.27) $\displaystyle\int_0^\pi x^2\sin x\,dx$ **1.28)** $\displaystyle\int_{e-1}^2 \log(x^2-x)\,dx\,dx$ **1.29)** $\displaystyle\int_1^e \log^3 x\,dx$

1.30) $\displaystyle\int_0^1 x^2\arctan x\,dx$ **1.31)** $\displaystyle\int_0^{\sqrt{\pi}} x^3\sin(x^2)\,dx$ **1.32)** $\displaystyle\int_0^{\pi/2} e^x\cos x\,dx$

1.33) $\displaystyle\int_0^{1/2} x\arcsin x\,dx$ **1.34)** $\displaystyle\int_0^{2\pi} x^2\sin^2 x\,dx$

1.35) $\displaystyle\int_2^3 \frac{x+1}{x+2}\,dx$ **1.36)** $\displaystyle\int_1^2 \frac{x-1}{x(x+1)}\,dx$ **1.37)** $\displaystyle\int_{-2}^3 \frac{x-8}{x^2-x-12}\,dx$

1.38) $\displaystyle\int_0^1 \frac{1}{x^3+2x^2-1}\,dx$ **1.39)** $\displaystyle\int_{-1}^2 \frac{2x+4}{x^2+3x+3}\,dx$ **1.40)** $\displaystyle\int_0^5 \frac{x^4+1}{x^2+x+2}\,dx$

1.41) $\displaystyle\int_{-1}^0 \frac{x^2-4x+1}{2x^3-x-1}\,dx$ **1.42)** $\displaystyle\int_2^3 \frac{x^3-x^2+x+2}{x^3-x}\,dx$ **1.43)** $\displaystyle\int_1^2 \frac{1+x}{x^4-2x^3+4x^2}\,dx$

2 Calcolare i seguenti integrali mediante opportune sostituzioni

2.1) $\displaystyle\int_0^1 \frac{e^x}{3-e^x}$ **2.2)** $\displaystyle\int_0^1 e^{\sqrt{x}}\,dx$ **2.3)** $\displaystyle\int_1^{e^\pi} \cos(\log x)\,dx$

2.4) $\displaystyle\int_0^1 \arctan(\sqrt{x})\,dx$ **2.5)** $\displaystyle\int_{-1}^1 \frac{1+\arctan x}{(1+\arctan^2 x)(1+x^2)}\,dx$

2.6) $\displaystyle\int_0^{\pi/4} \frac{1}{1+\sin^2 x}\,dx$ **2.7)** $\displaystyle\int_{-\pi/2}^{\pi/2} \frac{1}{1+\sin x+\cos x}\,dx$ **2.8)** $\displaystyle\int_{-\pi/6}^0 \frac{1+\tan x}{1-\tan x}\,dx$

2.9) $\displaystyle\int_0^{\pi/2} \frac{\sin(2x)}{2+\cos x}\,dx$ **2.10)** $\displaystyle\int_0^{\pi/2} \frac{1}{(1+\cos x)^2}\,dx$

2.11) $\displaystyle\int_1^2 \frac{\sqrt{1+x}}{x}$ **2.12)** $\displaystyle\int_0^1 \sqrt{9-x^2}\,dx$ **2.13)** $\displaystyle\int_0^1 \frac{2+\sqrt{x}}{1+\sqrt{x}}\,dx$

2.14) $\displaystyle\int_0^1 \sqrt{\frac{2x+3}{4x+2}}\,dx$ **2.15)** $\displaystyle\int_0^{\pi^2} \sin(\sqrt{x})\,dx$

3 Tracciare il grafico della funzione f e della corrispondente funzione integrale $F_0(x) = \displaystyle\int_0^x f(y)\,dy$

3.1) $f(x) = x-1$ **3.2)** $f(x) = x^2 - x$ **3.3)** $f(x) = \sin x$ **3.4)** $f(x) = e^{-x}$

3.5) $f(x) = 1+\cos x$ **3.6)** $f(x) = \dfrac{e^x}{1+e^x}$ **3.7)** $f(x) = \tan x$ (per $-\pi/2 < x < \pi/2$)

3.8) $f(x) = \begin{cases} 0 & \text{se } x \le 0 \\ 1 & \text{se } 0 < x \le 1 \\ 5 & \text{se } 1 < x \le 3 \\ -2 & \text{se } 3 < x \le 4 \\ 0 & \text{se } x > 4 \end{cases}$ **3.9)** $f(x) = \begin{cases} 0 & \text{se } x \le 0 \\ x & \text{se } 0 < x \le 2 \\ 4x-4 & \text{se } 2 < x \le 4 \\ 0 & \text{se } x > 4 \end{cases}$

3.10) $f(x) = \begin{cases} \cos(\pi x) & \text{se } x < -1 \\ x^3 & \text{se } -1 \le x \le 1 \\ \sqrt{x} & \text{se } x > 1 \end{cases}$ **3.11)** $f(x) = \begin{cases} \log|x| & \text{se } x \le -1 \\ e^{2x} & \text{se } x \ge -1 \end{cases}$

3.12) $f(x) = \sqrt{|x|}$ **3.13)** $f(x) = ||x-3|+1| - |x|$ **3.14)** $f(x) = [x]$

3.15) $f(x) = x - \sqrt[3]{x}$ **3.16)** $f(x) = ||x+1| - |x+2||$ **3.17)** $f(x) = \log(1+x)$

4 **4.1)** Dimostrare utilizzando la definizione di integrale che la seguente funzione $f : [0,2] \to \mathbb{R}$ è integrabile su $[0,2]$ con integrale nullo: $f(x) = \begin{cases} 0 & \text{se } x \neq 1 \\ 1 & \text{se } x = 1 \end{cases}$

4.2) Dimostrare che $\displaystyle\int_0^1 x\,dx = \frac{1}{2}$ utilizzando la definizione di integrale.

4.3) Sia $f : [0,1] \to \mathbb{R}$ continua su $[0,1]$ e tale che $f(x) \geq 0 \ \ \forall x \in [0,1]$. Sia inoltre $\displaystyle\int_0^1 f = 0$. Dimostrare che $f(x) = 0 \ \ \forall x \in [0,1]$.

4.4) Ricordando la formula trigonometrica $2\sin\alpha\cos\beta = \sin(\alpha+\beta) + \sin(\alpha-\beta)$, dati due numeri reali a, b calcolare l'integrale indefinito

$$\int \sin(ax)\cos(bx)\,dx.$$

Inoltre, dati due interi positivi n, m, dimostrare che $\displaystyle\int_0^{2\pi} \sin(nx)\cos(mx)\,dx = 0$.

4.5) Sia $f : [0,2] \to \mathbb{R}$ una funzione integrabile tale che $\displaystyle\int_0^2 f(x)\,dx = 4$;

- dimostrare che esiste $x_0 \in [0,1]$ tale che $f(x_0) < 3$;
- dimostrare che se f è continua su $[0,2]$ esiste $x_0 \in [0,2]$ tale che $f(x_0) = 2$; dimostrare inoltre che non si può giungere alla stessa conclusione senza l'ipotesi di continuità di f.

5 Calcolare l'area dei seguenti insiemi nel piano

5.1) $\{(x,y) \in \mathbb{R}^2 : y \leq x^3,\ y \leq 2-x,\ y \geq 0,\ x \geq 0\}$ **5.2)** $\{(x,y) \in \mathbb{R}^2 : x^3 \leq y \leq \sqrt[3]{x}\}$

5.3) $\{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 1-x^2\}$ **5.4)** $\{(x,y) \in \mathbb{R}^2 : 0 \leq y \leq 2-x^2,\ y \geq x\}$

5.5) $\{(x,y) \in \mathbb{R}^2 : 1 \leq x \leq 3,\ 0 \leq y \leq \frac{1}{x}\}$ **5.6)** $\{(x,y) \in \mathbb{R}^2 : \frac{1}{x} + x - 3 \leq y \leq 0,\ x \geq 0\}$

5.7) $\{(x,y) \in \mathbb{R}^2 : \sqrt{3}x \leq y \leq \sqrt{1-x^2},\ x \geq 0\}$ **5.8)** $\{(x,y) \in \mathbb{R}^2 : |x| \leq y \leq 1-x^2\}$